CORNELL UNIVERSITY
Mechanical and Aerospace Engineering

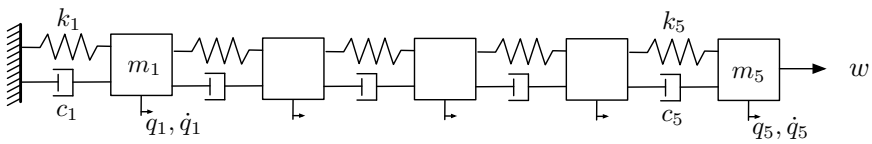
M&AE 6760: Model Based Estimation

Prof. M. Campbell | **Homework #3** | Date: 1 Mar 2026
Spring 2026 | Due: (5:00pm) 13 Mar 2026

Note that suggested MATLAB *.m commands are given in some problems. These are only suggestions, as there may be other commands or Python equivalents you could use.

Problem #1: Multi-State KF with different measurements

This problem explores the ability of a Kalman Filter to estimate states for different measurements. Consider a flexible system (similar to a cantilevered rod) modeled as an inter-linked give mass-spring-damper system:



MATLAB and Python scripts are given on Canvas with baseline models, simulation and plotting. You can modify the code for each sub-problem at locations denoted as %% YOUR CODE HERE. For the most part, you probably will not have to change the plotting functions, but you may have to zoom in to see different aspects of the estimator performance. For MATLAB, you can 'Run Section' for each part. For the solutions, please include the figures with your discussion and append your code as an appendix.

Please answer the following:

a) Assume process noise w is acting (see m-file) and there is an initial displacement of the fifth mass. Create and run two Kalman Filters, one using the measurement $z_{q1} = q_1 + v_{q1}$ and the other $z_{q5} = q_5 + v_{q5}$. Choose (and justify via discussion) initial conditions that you believe are 'reasonable.' Compare/discuss the ability of the filters to estimate the position/velocity of each mass.

b) Consider the same Kalman Filter from (a) using measurement $z_{q5} = q_5 + v_{q5}$. Plot the 'predicted' $(k+1|k)$ and 'updated' $(k+1|k+1)$ estimator variance of the 5th mass, i.e. $\sigma^2_{q_5} = E[(q_5 - \hat{q}_5))^2]$. Discuss the trends in how the two lines compare, and how these trends fit with the prediction and update steps of the Kalman Filter. You may find using semilogy.m makes it easier to evaluate.

c) For the same Kalman Filter from (a) using the measurement $z_{q5} = q_5 + v_{q5}$, calculate the steady state predicted error covariance $\bar{P}$; in MATLAB, you will use idare.m (see KF2 lecture). Using $\bar{P}$, consider the variance of the 5th mass $\sigma^2_{q_5} = E[(q_5 - \hat{q}_5))^2]$. How does this value relate to results/plots in (b)?

d) Consider the same Kalman Filter from (a) for the measurement $z_{q5} = q_5 + v_{q5}$, and the steady state predicted error covariance $\bar{P}$ and Kalman gain $\bar{K}$ from (c). Design a Kalman Filter using $\bar{P}$ as the initial error covariance and use the steady state Kalman gain $\bar{K}$ for the filter update. Compare and discuss the performance of this filter with that from (a).

e) Consider the case of a closed loop controller given by:

$$u = -K_z z$$

where $z = z_{q5}$ and K_z is a controller constant. Find the closed loop state space model which would be used for a Kalman Filter to estimate the full state. How is this model different than the open loop model? Will these model differences change the implementation of the Kalman Filter?

Problem #2: **Kalman Filter and Integrity**

Consider an aircraft moving in a plane with constant speed (i.e. magnitude of velocity) and turning with a constant angular rate. Such a model is often used by air traffic control tracking algorithms to describe an aircraft executing coordinated turns. Given 2D inertial position variables $\xi(t)$ (East position) and $\eta(t)$ (North position), the equations of motion are

$$\ddot{\xi} = -\Omega\dot{\eta} + F_\xi$$
$$\ddot{\eta} = \Omega\dot{\xi} + F_\eta$$

where Ω is the constant angular rate, such that $\Omega > 0$ implies a counterclockwise turn. Using the state representation $x(t) = [\xi, \dot{\xi}, \eta, \dot{\eta}]^T$, it can be shown that the discrete state matrix is

$$F_\mathsf{D} = e^{F_\mathsf{C}\Delta t} = \begin{bmatrix} 1 & \frac{\sin(\Omega\Delta t)}{\Omega} & 0 & -\frac{1-\cos(\Omega\Delta t)}{\Omega} \\ 0 & \cos(\Omega\Delta t) & 0 & -\sin(\Omega\Delta t) \\ 0 & \frac{1-\cos(\Omega\Delta t)}{\Omega} & 1 & \frac{\sin(\Omega\Delta t)}{\Omega} \\ 0 & \sin(\Omega\Delta t) & 0 & \cos(\Omega\Delta t) \end{bmatrix}$$

where F_C is the CT LTI state matrix for this system.

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Please complete the following:

a) Given 2D position measurements of one aircraft, develop a Kalman Filter estimating the full state. Plot the 2D position error over time (error is easier to visualize), with the estimator 2σ uncertainties and discuss.

b) Given the same Kalman Filter as in (a), set up a 95% validation gate on the measurements. You do not have to reject the measurements - just measure whether they pass the gate. Note that a addition to the plot_estimator is given to circle the errors that are rejected. How many measurements are rejected? How do the number of measurements rejected compare to the formal statistics of the hypothesis test? Discuss qualitatively the measurements are rejected from the time history plot - are the measurements always outside the 2σ bounds?

c) Download the ACdata.mat file from Canvas. This file contains measurement data for a flight with different wind/noise conditions. You are to develop a Kalman Filter similar to (a), and use the filter integrity to tune the noise covariance Q. Note that a plotting routine for the test statistics is given - be sure to match the naming and structure of the variables.

d) Consider now two aircraft, both with 2D position measurements. Create a single Kalman Filter with four measurements: two position measurements for each aircraft. The MATLAB function blkdiag.m is useful here. How do the the results from aircraft A compare to those from (a)? What is unique about the covariance matrix?

e) Repeat (d), but using only a relative range measurement between the aircraft, given as

$$\mathbf{z}_{\text{range}} = \begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x}^A \\ \mathbf{x}^B \end{bmatrix} + \mathbf{v}_{\text{range}}$$

How well does this filter work compared to (d)? Why is is better or worse?